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MILITARY SPECIFICATION

TUBING, CHROME-MOLYBDENUM, 4130 STEEL, SEAMLESS AND WELDED, AIRCRAFT QUALITY

This specification is mandatory for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers chromium-molybdenum (4130) seamless and welded steel tubing of aircraft quality.

1.2 *Classification.* The tubing shall be of the following grades, types, and physical conditions, as specified (see 6.2):

Grades:

All tubing—Regular grade, except that type I, condition (N) tubing is also available in a special (S) grade having reduced decarburization limits.

Types:

Type I—Round
Type II—Rectangular or square
Type III—Streamline
Type IV—Oval

Physical conditions:

(A)—Annealed
(N)—Normalized or stress relieved
(HT-125)—Heat treated to a minimum tensile strength of 125,000 psi
(HT-150)—Heat treated to a minimum tensile strength of 150,000 psi
(HT-180)—Heat treated to a minimum tensile strength of 180,000 psi

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification of the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-I-6868—Inspection Process, Magnetic Particle
MIL-C-16173—Corrosion Preventive Compound, Solvent Cutback, Cold Application

STANDARDS

FEDERAL

Fed. Test Method Std. No. 151—Metals Test Methods
Fed. Std. No. 183—Continuous Identification Marking of Iron and Steel Products

MILITARY

MIL-STD-129—Marking for Shipment and Storage
MIL-STD-163—Steel Mill Products Preparation for Shipment and Storage
MS33529—Tolerances—Seamless Steel Tubing, Aircraft Airframe, Round
MS33532—Square and Rectangular Tubing—Carbon Steel and Alloy Steel, 0.3 Carbon, Maximum
MS33530—Tolerances—Welded Carbo and Alloy Steel Tubing
MS33534—Standard Dimensions fc Streamline and Oval Tubular Shapes

AIR FORCE-NAVY AERONAUTICAL

AND10102—Tubing—Standard Dimer sions for Round Seamless Alloy Steel

(Copies of specifications, standards, drawings, ar publications required by suppliers in connection wi specific procurement functions should be obtained fro the procuring activity or as directed by the contractir officer.)

2.2 Other publications. The following dor uments form a part of this specification to tl extent specified herein. Unless otherwise i dicated, the issue in effect on date of invitatic for bids or request for proposal shall apply:

FSC 4710

770-868—65

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SOCIETY OF AUTOMOTIVE ENGINEERS AEROSPACE MATERIAL SPECIFICATIONS

AMS 2301—Aircraft Quality Steel Cleanliness, Magnetic Particle Inspection Procedure

AMS 2640—Magnetic Particle Inspection

(Copies of SAE publications may be obtained from the Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, N.Y., 10017.)

3. REQUIREMENTS

3.1 Data. Unless otherwise specified in the contract or order, no data are required by this specification or any of the documents referenced in section 2 (see 6.2).

3.2 Material. The tubing shall be of aircraft quality. The material shall be magnetically inspected in accordance with the procedures of AMS 2301, and shall not exceed the size and frequency rating limits indicated in the paragraph entitled "Disposition" of AMS 2301 (see 4.5).

3.2.1 *Manufacturing process.* The steel shall be manufactured by the open-hearth or electric-furnace process, unless a single process is specified in the contract or purchase order.

3.2.1.1 Sufficient discard shall be taken from each ingot to ensure freedom from piping and undue segregation.

3.3 Chemical composition. The chemical composition of tubing shall be as specified in table I.

TABLE I. *Chemical composition*

Element	Analysis, percent	Check analysis tolerance,[1] percent
Carbon	0.27–0.33	±0.02
Manganese	.40– .60	±.03
Phosphorus	.025 (max)	+.005
Sulfur	.025 (max)	+.005
Silicon	.20–0.35	±.02
Chromium	.80–1.10	+.05, −0.03
Molybdenum	.15–0.25	±.02

[1] Individual determinations may vary from the specified range to the extent shown in the check analysis column, except that elements in any heat shall not vary both above and below the specified range.

3.4 Mechanical properties. After the last cold-draw pass, the tubing shall be normalized, stress relieved, or otherwise heat treated to develop the mechanical properties specified in table II.

3.5 Dimensions. The dimensions of type I tubing shall conform to the standard dimensions shown on AND10102. The dimensions of types III and IV shall conform to the standard dimension shown on MS33534. The dimensions

TABLE II. *Mechanical properties*

Condition and wall thickness	Tensile strength (min.)	Yield strength at 0.2 percent set or at extension indicated		Elongation in 2 inches	
		(Min.)	Extension under load	Full tube (min.)	Strip (min.)
Inch	psi	psi	Inches in 2 inches	Percent	Percent
(A)	[1] 95,000				
(N)					
Up to 0.035 incl	95,000	75,000	0.0090	10	5
Over 0.035 to 0.187 incl	95,000	75,000	.0090	12	7
Over 0.187	90,000	70,000	.0087	15	10
(HT-125) All walls	125,000	100,000	.0107	12	7
(HT-150) All walls	150,000	135,000	.0130	10	6
(HT-180) All walls	180,000	165,000	.0154	8	5

[1] Maximum.

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of type II tubing shall be as negotiated between the supplier and purchaser.

3.6 Length.

3.6.1 *Exact lengths.* Tubing of all sizes may be ordered to exact lengths or in lengths expressed as a multiple of a definite unit, with tolerances as specified in the contract or purchase order.

3.6.2 *Mill lengths.* When exact or multiple lengths are not specified (see 6.2), tubing will be accepted in mill lengths of 5 to 20 feet, but not more than 10 percent of any order shall be furnished in lengths shorter than 12 feet.

3.7 Tolerances.

3.7.1 *Type I.* The permissible variations in dimensions of type I seamless tubing shall be as shown on MS33529 and for type I welded tubing as shown on MS33530.

3.7.2 *Types II, III, and IV.* The permissible variations in dimensions of all types II, III, and IV tubing shall be as shown on MS33532.

3.8 Grain size. The austenitic grain size of the steel used for this tubing shall be predominantly No. 5 or finer, with grains as large as No. 3 permissible, as determined by the method specified in section 4. For seamless tubing, grain size shall be determined on a billet before piercing, hot working, or cold drawing.

3.9 Decarburization.

3.9.1 *Regular grade.* The average depth of total decarburization of regular grade, condition (N) tubing, as received from the tubing manufacturer, shall not exceed the amounts listed in table III. The depth of decarburization in table III is the total decarburization, or the sum of the depths of complete decarburization and partial decarburization on both the inner and outer surfaces of the tube. No more than 75 percent of the amount listed in table III is to appear on the outer surface of the tube. The depth of complete decarburization shall not exceed one-half the allowable total decarburization. (The word "average" means the results of several readings on a cross section of tubing.)

3.9.2 *Special grade.* The permissible decarburization of (S) grade, type I condition (N) tubing shall be one-half that listed in table III, with no complete decarburization permissible.

TABLE III. *Decarburization limits*

Nominal (wall) thickness (inch)	Allowable total decarburization (ID+OD) (inch)	Maximum total decarburization (OD) (inch)
Up to 0.040	0.010	0.008
0.041 to 0.050	.012	.009
0.051 to 0.070	.014	.011
0.071 to 0.080	.016	.012
0.081 to 0.090	.018	.014
0.091 to 0.100	.020	.015
0.101 to 0.150	.022	.017
0.151 to 0.200	.026	.020

3.10 Condition. Unless otherwise specified, tubing shall be furnished in regular grade, type I, condition (N).

3.11 Identification of product. Each tube shall be marked in accordance with Fed. Std. No. 183 and shall include the number of this specification.

3.12 Workmanship. The tubing shall have a finish conforming to the best practice for aircraft quality material. It shall be smooth, clean, and free from heavy scale or oxide on the interior and exterior surfaces, and shall be free from burrs, seams, tears, grooves, laminations, slivers, pits, and other injurious defects. Surface imperfections, such as handling marks, straightening marks, light mandrel and die marks, shallow pits, and scale pattern will not be considered as injurious defects, provided the imperfections are removable without reducing the diameter or wall thickness of the tubing below the permissible tolerance limits. The removal of surface imperfections is not required.

3.12.1 *Weld.* The weld shall not contain defects greater than $\frac{1}{8}$-inch long or depths greater than one-half the wall thickness. The welded tubing may be pickled or otherwise cleaned, if necessary, to meet surface conditions specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase

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order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. The examination and testing of tubing shall be classified as quality conformance inspection.

4.2.1 Sampling and inspection shall be performed in accordance with Federal Test Method Standard No. 151 and as specified herein. If the material is taken from stock and is not identifiable as to heat and method of manufacture, or if the identity of any portion of the shipment is obscure in any respect, additional samples shall be selected to determine conformance of all portions of the shipment to this specification.

4.3 Lot. A lot shall consist of tubing produced from the same heat and which is essentially homogeneous in all respects, in the same condition, of the same type, size, and wall thickness offered for delivery at the same time.

4.4 Examinations.

4.4.1 *Examination of product.* Each length of tubing shall be visually examined for compliance with surface condition and workmanship requirements. Samples selected in accordance with table IV shall be examined to assure compliance with the specified dimensions and tolerances, and identification marking requirements.

TABLE IV. *Sampling plan*

Lot size	Sample size	Acceptance number rejectable
1 to 15	All	0
16 to 180	15	0
181 to 300	35	0
301 to 500	50	1
Over 500	75	2

4.4.2 *Examination of preparation for delivery.* Preparation for delivery shall be examined for conformance to section 5.

4.5 Magnetic inspection. Specimens shall be selected, inspected, and rated in accordance with the procedures of AMS 2301. Inspection shall be in accordance with MIL–I–6868 or AMS 2640.

4.6 Chemical analysis.

4.6.1 *Sampling.* At least one sample shall be selected for check chemical analysis in accordance with Federal Test Method Standard No. 151 to represent each heat in the lot. The sample shall consist of not less than 2 ounces.

4.6.2 *Method.* Specimens shall be prepared in accordance with Methods 111 or 112 of Federal Test Method Standard No. 151, and shall be tested by wet chemical or spectrochemical methods. In the event of dispute, analysis shall be by wet chemical methods.

4.6.3 *Waiver.* Samples for check of chemical analysis may be waived provided that all of the material in the lot can be identified as being made from a heat previously analyzed and found to conform to the chemical composition specified herein.

4.7 Tensile test.

4.7.1 *Sampling.* At least one tensile test sample shall be selected from each 1,000 feet or less of each lot for determination of mechanical properties.

4.7.2 *Preparation of specimens.* Tensile test specimens shall be prepared in accordance with Method 211 of Federal Test Method Standard No. 151.

4.7.3 *Method.* Tensile tests and determination of yield strength and elongation shall be conducted in accordance with Method 211 of Federal Test Method Standard No. 151.

4.8 Grain size.

4.8.1 *Sampling.* One or more samples shall be selected from one or more billets used in making the tubing and suitable for determining the austenitic grain size.

4.8.2 *Preparation of specimens.* The specimens shall be taken one-half way between the center and outside of the billet. The specimens shall be approximately 1-inch square or round, and normalized at 1,650° F.

4.8.3 *Method.* The grain size shall be determined by procedure B or D of Method 311 of Federal Test Method Standard No. 151.

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4.9 *Decarburization.*

4.9.1 *Sampling.* At least one cross sectional sample shall be selected from each 1,000 feet or less of each lot. The sample shall be cut at least three-fourths of an inch from the mill end.

4.9.2 *Preparation of specimens.* The specimens shall be metallographically prepared and etched in 5 percent nital.

4.9.3 *Method.* The specimens shall be examined at 100 diameters.

4.10 Rejection and retest. Failure of a specimen to meet the test requirements shall be cause for rejection of the lot. At the discretion of the contractor or supplier, or both, retest will be permitted. A retest sample of five specimens shall be tested to replace each failed specimen of the original sample. If one of the retest specimens fail, the lot shall be rejected with no further retesting permitted.

5. PREPARATION FOR DELIVERY

5.1 *Preservation. Unless otherwise specified,* all tubing shall be coated inside and outside with corrosion-preventive compound conforming to MIL-C-16173, which shall be diluted by the addition of 50 percent, by volume, of the solvent specified in paragraph entitled "Solvent" of MIL-C-16173.

5.2 Preparation for shipment. Unless otherwise specified, tubing shall be prepared for shipment in accordance with level A of MIL-STD-163 (see 6.2).

5.3 *Marking of shipment.* Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tubing is intended for use in aircraft construction, particularly where parts are fabricated by welding. Preheating prior to welding is desirable. The special grade tubing is preferable for applications involving bending fatigue loads inasmuch as the tubing surface is of higher strength than the surface of regular grade tubing.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Size and wall thickness (for round cross sections, see AND10102).
(c) Amount (aggregate length in feet; length of each piece when exact lengths are desired) (see 3.6).
(d) The type, if type I is not desired (see 1.2).
(e) The condition and grade if condition (N) regular grade is not desired (see 1.2 and 3.10).
(f) Seamless or welded, as desired.
(g) Data requirements (see 3.1).
(h) Level of packaging, if other than level A (see 5.2).
(i) Preservation, if other than specified in 5.1.

6.2.1 *Mechanical properties.* When tubing is to be heat treated (quenched and drawn) to definite mechanical properties specified by the procuring activity, the contract or order should state the desired mechanical properties, and the manufacturer should establish the optimum quenching range and quenching medium to be used.

Custodians:
Army—MR
Navy—WP
Air Force—(71)
DSA—CS
Reviewer activities:
Army—MR, MI
Navy—WP
Air Force—(11), (69)
User activities:
Army—MO
Navy—
Air Force—
Preparing activity:
Air Force—(11)
Project No. 4710-0108

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SPECIFICATION ANALYSIS SHEET

FORM APPROVED BUDGET BUREAU NO. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-T-6736B Tubing, Chrome-Molybdenum, 4130 Steel, etc., AC Quality

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *IF "YES" IN WHAT WAY?*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Commander, Systems Engineering Group

Attn: SEPS

Wright-Patterson Air Force Base, Ohio 45433

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